

SelectiveSR: A new Gaussian Splatting super-resolution strategy

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Abstract

Image super-resolution remains an important problem for computer vision optimisation, as it produces a higher quality image with greater visibility. Arbitrary-scale super-resolution is the type of super-resolution in which the process can happen using only one model, allowing the user to toggle different scales as they wish. Recent research in this area has uncovered the promise of Gaussian Splatting in super resolution, revealing its strength in producing a more complete representation of the original image through a continuous distribution space, as opposed to a discrete one. In particular, the ContinuousSR and GaussianSR frameworks have shown promising results, yielding high-quality images and impressive PSNR scores across a diverse range of images. However, ContinuousSR can struggle with achieving a sufficient level of precision at the local level, limited by its global Deep Gaussian Prior module. In addition, GaussianSR can struggle with the reverse problem, lacking a representative global kernel selector. Here, we introduce a new arbitrary-scale super-resolution framework that combines the best of these two strategies, merging the tunable Gaussian kernel selection bank from GaussianSR with the global prior from ContinuousSR. The result is that we maintain competitive PSNR and SSIM performance with four state-of-the-art baselines on the standard comprehensive Div2k dataset, while achieving at least 2x higher computational efficiency as measured by testing time. Overall, these results underscore the necessity for a unified hybrid approach to Gaussian splatting-based super-resolution, providing a promising new framework for producing higher-quality-looking images.

1. Introduction

“Since its introduction last year, 3D Gaussian Splatting has helped drive a renaissance in 3D as-

set and content creation.”

- Amazon Web Services Spatial Computing (2024 August 27) [6]

Over the span of just the previous few decades, the world has seen a massive increase in the use of smartphones for taking photos, with phone cameras and cameras in general experiencing a tremendous improvement in the level of convenience and quality of the images. Despite these positive changes, however, the ideal camera remains an open problem, as some cameras like the ones at signals or public places can still struggle with producing low-resolution (LR) images. There are numerous potential explanations for this reduced quality such as the quality of the sensor embedded within the camera or limitations on final image size as a requirement, and computer vision research has proposed many effective solutions thus far.

In particular, recent work has found promise in **arbitrary-scale super-resolution** (ASSR) and in **Gaussian Splatting**. To understand ASSR, we can break it down word by word. Super-resolution is acquiring a high-resolution (HR) image from an LR image, thereby turning the LR image into something of higher clarity and quality. Then, “arbitrary-scale” is upscaling the LR image into the desired resolution of the user. This is in contrast to traditional super-resolution approaches that could only build one model for some fixed image scaling, such as 2x or 3.3x. ASSR allows the use of just one model to perform any level of image scaling.

On the other hand, **Gaussian Splatting** (GS) is a three-dimensional (3D) rendering and scene manipulation strategy, and it has been adapted to the two-dimensional (2D) space of images by using 2D Gaussian distributions on precise scene projections. The result is that we now have continuous field distributions rather than discrete data points, allowing better representation of the image pixels. In particular, GS affords a better representation of the radiance field of the pixel, which is the distribution of light that re-

flects off of the pixel. Being a continuous field, this space of light pixels cannot be captured as well by a discrete number distribution. Instead, a continuous number distribution is the natural representation. [8]

Because of this stronger representation of the image pixel space, super resolution approaches have evolved towards its greater use, finding new variations on the method to create higher-quality super-resolved images. In particular, we highlight three of these strategies in our project, with each one solving a specific subproblem within the goal of super-resolution. **GaussianImage** solves the problem of too many parameters by eliminating the 3D parameters, and replacing them with 2D Gaussians that are represented by a similar set of variables: position, covariance, colour, and opacity [10]. This slashes the memory overhead. **GaussianSR** achieves a Gaussian field representation for each feature vector by blending sampled Gaussian kernels from a bank via a selective kernel method called Selective Gaussian Splatting (SGS), increasing the accuracy for the interpolation of each pixel in the final image [4]. Last, **ContinuousSR** optimizes the super-resolution process by using Gaussian kernels from a distribution known as the Deep Gaussian Prior distribution. These kernels are further optimized using a parallel process called Adaptive Position Drifting, which fine-tunes the spatial coordinates of these kernels in the final image [2].

Here, our goal is to develop a Gaussian Splatting-based super-resolution framework that combines the global statistical consistency of ContinuousSR with the local specialization of SGS. At a high level, our approach integrates principles from both architectures to produce a model that is globally coherent, locally adaptive, and capable of generating high-resolution images at arbitrary scales. This positions our method as a step toward unifying the complementary strengths of existing Gaussian-based SR strategies.

In Section 2, we provide a deeper overview of the three foundational works in Gaussian Splatting super-resolution that were introduced above, along with several other important state-of-the-art (SOTA) works. In Section 3, we describe the novel idea for our strategy in detail. In Section 4, we present the experiments that we perform on the framework, describing at length the setup and the results, the reasons behind the performance differences between SelectiveSR and the SOTA, and the future possible explorations from our work.

2. Related Works

SR has been performed using several SOTA strategies within recent years. **Bicubic interpolation** [5] is a naïve approach which computes each super-resolved pixel via a polynomial function composed of the four nearest “corner” pixels around the new pixel. The image is organised into a grid of squares, with a pixel belonging to a corner of a

square, so that for each pixel in the new image, there are four surrounding corners which represent pixels. Due to its limited dependence on only these surrounding pixels, bicubic interpolation tends to fail on more complex images, producing blurs or other deformities to the final image. **Local Implicit Image Function (LIIF)** [3] is a more advanced approach, touching into ASSR via training a multilayer perceptron to learn a mapping from the original image features to RGB values. As a result, the framework can take on any given scale that the user desires, making it an ASSR strategy. **Implicit Transformer Super-Resolution Network (ITSRN)** [9] expands upon LIIF to change the mapping from RGB to RGB to instead coordinate to RGB, with the idea being that the pixel features and coordinates are different and thus cannot be used together as they are in LIIF. While these methods have shown great progress in traditional SR, recent work has shifted toward continuous image representations, specifically based on Gaussian Splatting.

One of these earlier works was **GaussianImage** [10], which introduced the idea that a 2D image can be represented as a continuous field of Gaussians rather than a discrete pixel grid. Its main contribution is an accumulated-blending renderer, which offers a simpler way to blend Gaussians. Instead of sorting them by depth, the renderer combines them using a clean weighted sum, enabling real-time performance (1500–2000 FPS). Although GaussianImage was not inherently developed for super-resolution, it established two principles that later SR methods rely on: (1) images can be expressed through explicit Gaussian primitives parameterized by position, covariance, and color, and (2) these continuous fields can be evaluated efficiently at arbitrary coordinates. These insights form the representational foundation upon which Gaussian-splatting-based super-resolution models were built.

GaussianSR [4] extends this concept from image space into feature space. Instead of treating LR features as discrete points, GaussianSR models each feature vector as a small continuous Gaussian field. Its central mechanism, **Selective Gaussian Splatting (SGS)**, assigns each pixel one kernel from a trainable Gaussian kernel bank, enabling region-specific behavior: sharp kernels for textures and broader kernels for smooth regions. This makes GaussianSR a local SR method — powerful for specialized textures but limited by the fixed kernel bank and the absence of global statistical constraints. It also retains the sequential, multi-step structure of encoder → Gaussian refinement → decoder, making it slower than more recent approaches like ContinuousSR.

Another such model built upon the GaussianImage concept is **ContinuousSR** [2]. It is a GS approach that achieves rapid ASSR through the parallel deployment of three separate units, each one devoted to a specific generation. One of these units is **Deep Gaussian Prior-Driven Covariance**

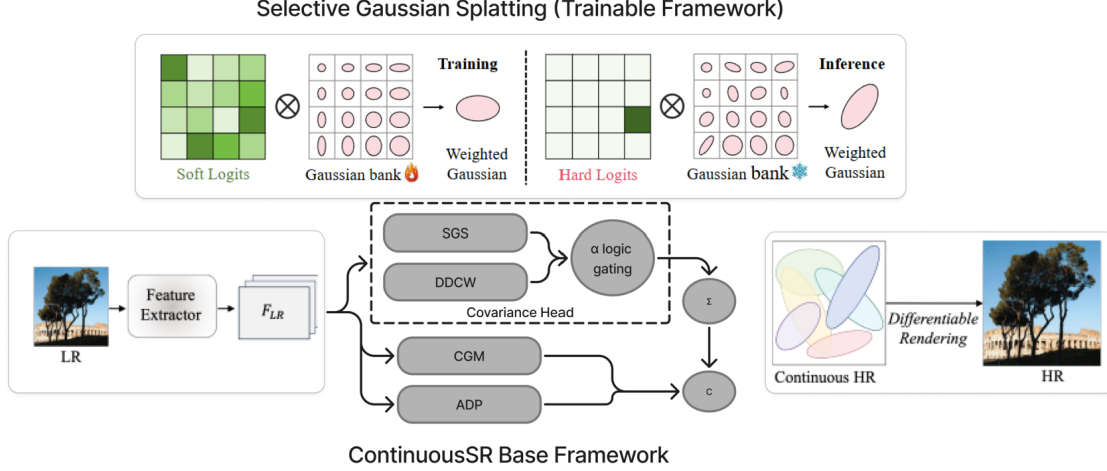


Figure 1. **Overview of the SelectiveSR architecture.** Our model extends the ContinuousSR framework by introducing a hybrid covariance head that fuses a global DDCW-based covariance with a locally specialized SGS-based covariance via a learned per-pixel gating weight α . APD and CGM remain unchanged from the original pipeline. The resulting Gaussian field is rendered differentially to obtain the high-resolution image.

Weighting (DDCW), which produces Gaussian kernels that are then rendered into the final HR image. The idea in DDCW is to use prior knowledge, which we express via a predefined Gaussian distribution called Deep Gaussian Prior (DGP). The module first samples the covariance parameters σ_x^2 , σ_y^2 , and ρ from this DGP to form a dictionary of Gaussian kernel types, which are then combined via a weighted sum to produce the desired Gaussian kernel. These weights come from an adaptive module M_{weight} that is based on the LR feature maps. Another one of the units in ContinuousSR is Adaptive Position Drifting (APD), the idea of which is to tune the positions of the final pixels. This is needed since the idea behind super resolution is to map each LR pixel to multiple HR pixels, which can introduce spatial variance in the pixel. Lastly, the third unit is Colour Gaussian Mapping (CGM), which predicts the RGB colours of the final pixels. There are two areas of novelty in ContinuousSR. One area of novelty is two of the units of ContinuousSR, DDCW and APD. The other area of novelty is the efficiency enhancement, as previous strategies relied on repeated iterations of the same process as opposed to ContinuousSR, which needs only one pass. Because of the competitive performance of ContinuousSR with regards to nine other SOTA works and its superior computational efficiency as measured by the GPU memory usage, runtime, and FLOP count, we focus on creating a new technique that borrows elements from this architecture.

Despite the impressive efficiency and global consistency of ContinuousSR, one limitation remains: its Gaussian generation mechanism, Deep Gaussian Prior-Driven Covariance Weighting, produces covariance kernels using a global prior, the Deep Gaussian Prior. While this yields stable,

globally coherent kernels, it lacks the local adaptivity required to handle fine-grained, high-frequency textures. For example, datasets such as Manga109 contain extremely sharp line art, thin strokes, and repetitive hatch patterns that require highly specialized, narrow kernels to avoid over-smoothing. In contrast, GaussianSR’s SGS module excels at exactly this kind of localized specialization by selecting a content-appropriate kernel from a learned bank. However, SGS suffers from slower sequential processing and lacks the global statistical regularization provided by ContinuousSR. This exposes a fundamental gap in current Gaussian-splatting SR frameworks: no existing method unifies global statistical consistency, local specialization, and fast inference within a single architecture.

To address this gap, we introduce **SelectiveSR**, a hybrid Gaussian-splatting framework that combines the global covariance structure of ContinuousSR with the locally adaptive kernel behavior of GaussianSR. A detailed description of our method is provided in Section 3.

3. Methodology

Overview. Our model adopts the direct-to-signal idea from ContinuousSR: instead of predicting an intermediate feature map and decoding it, the network takes a low-resolution (LR) image and directly outputs a 2D Gaussian field that can be rendered at any resolution. We keep the original *Adaptive Position Drifting* (APD) and *Color Gaussian Mapping* (CGM) modules intact, and we focus on modifying how Gaussian covariances are produced.

To address the gap identified in Section 2, we introduce a *hybrid covariance head*. ContinuousSR generates globally consistent covariances using the DDCW module,

while GaussianSR specializes covariances locally through a learned kernel bank. Our hybrid head combines both sources: it computes a global covariance from DDCW, a locally adaptive covariance from an SGS-style branch, and blends them using a learned per-pixel gating scalar $\alpha \in [0, 1]$. This allows each pixel to choose, dynamically, how much to rely on global structure versus local specialization. The formal definition of this blending is provided in Section 3.2.

An overview of the full architecture, including the hybrid covariance head and its integration into the ContinuousSR pipeline, is shown in Fig. 1.

3.1. Hybrid Covariance Head

The hybrid head produces two covariance matrices for each pixel location. The DDCW branch follows ContinuousSR exactly, predicting a weighted combination of DGP-sampled kernels to form Σ^{DDCW} . In parallel, the SGS branch predicts logits over a trainable kernel bank, which are converted into a soft assignment during training (via Gumbel–Softmax) and a hard kernel selection during inference. This is similar to GaussianSR, yielding Σ^{SGS} . Both matrices are parameterized to ensure positive semi-definiteness.

We modify only the covariance generation pathway; APD and CGM remain unchanged. This design choice reflects our observation that *covariance geometry*—the orientation, anisotropy, and spread of the Gaussian kernel—is the primary factor governing detail preservation in arbitrary-scale super-resolution. The structure of this module is shown in the upper portion of Fig. 1.

3.2. α -gating Logic

To blend the two covariance estimates, the model predicts a gating value α_i for each pixel i . We implement this using a lightweight 1×1 convolution applied to the shared feature map. This produces an $m \times n$ map of gating values for an $m \times n$ LR image. A sigmoid activation ensures $\alpha_i \in [0, 1]$.

Given these values, the covariances are combined according to Eq. 1:

$$\Sigma_i = \alpha_i \Sigma_i^{\text{SGS}} + (1 - \alpha_i) \Sigma_i^{\text{DDCW}}. \quad (1)$$

The gating mechanism allows the network to decide, pixel by pixel, whether global DGP-informed smoothing or local SGS specialization is more appropriate. Empirically, regions with sharp edges, thin lines, or structured textures tend to produce higher α , while flatter or low-frequency regions rely more on the global estimate.

4. Experiments

4.1. Setup

Datasets. We use a standard image super-resolution training dataset and validation dataset, **Div2K** [1]. Div2K is made up of 1000 different 2K-resolution images spanning humans, urban areas, and flora and fauna, making for an effective evaluation of the versatility of the method.

Training configuration. To train, we use the training configuration found from the three highlighted SOTA Gaussian strategies: Adam optimiser, $1e-4$ learning rate with a 0.5 decay rate for 20 epochs, 100 epochs, L1 loss, and batch size 1-2. For the performance evaluation metrics, we use the standard peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM). For the efficiency evaluation metrics, we measure training time (seconds), number of parameters, and graphical processing unit (GPU) memory utilization. For the baselines, we compare our method against BiCubic, LIIF, ITSRN, and GaussianSR.

4.2. Results

Performance. We evaluate the quantitative performance of our proposed model against the baselines on the Div2K dataset. Table 1 summarizes the training and testing PSNR and SSIM scores.

Our model achieves the highest PSNR on the testing set (30.0299 dB), outperforming both the traditional BiCubic baseline and the recent GaussianSR method (29.7521 dB). While our training metrics matched GaussianSR exactly, our method demonstrated superior generalization in PSNR during testing. However, we observed a trade-off in the structural similarity index (SSIM) during testing (0.8198), which suggests areas for future refinement in texture preservation compared to the baselines.

Table 1. Quantitative Performance Comparison (PSNR and SSIM) on Div2K

Model	Train PSNR	Test PSNR	Test SSIM
BiCubic	26.7873	26.7873	0.7707
LIIF	29.4172	28.5835	0.8242
ITSRN	29.8398	28.9633	0.8334
GaussianSR	35.4815	29.7521	0.8134
Our Model	35.4815	30.0299	0.8198

Efficiency Analysis. We further analyzed the computational efficiency of our method, as detailed in Tables 2 and 3. Our model demonstrates significant improvements in memory usage and inference speed. It utilizes only 8 GB of GPU memory during training, which is comparable to LIIF and significantly lower than the 12 GB required by GaussianSR.

Most notably, our model achieves the fastest inference runtime (2.1 minutes) among the learning-based methods, roughly 2.5 times faster than GaussianSR (5.2 minutes), over 11 times faster than ITSRN (23.1 minutes), and over 8 times faster than ITSRN (16.5 minutes). This speed is achieved despite a larger parameter count (44.9 M), indicating that our architecture optimizes the forward pass efficiently despite the model size.

Table 2. Efficiency Comparison: Memory, Runtime, and Model Size

Model	Train Memory	Train Time	Test Time
BiCubic	-	-	6.6 min
LIIF	8 GB	2.2 hr	16.5 min
ITSRN	10 GB	6 hr	23.1 min
GaussianSR	12 GB	2.0 hr	5.2 min
Our Model	8 GB	10.0 hr	2.1 min

Table 3. Efficiency Comparison: Model Size

Model	Parameters
BiCubic	0
LIIF	1.57 M
ITSRN	22.6 M
GaussianSR	1.8 M
Our Model	44.9 M

Ablation. Finally, we perform ablations on our framework. Because our method is a new expansion of prior existing works, we present our ablation in the form of the GaussianSR test results on the Div2K dataset, which were shown in Table 1. As shown by these results, our technique remains competitive with this previous work, with even a slight performance superiority over it. At the same time, we achieve higher efficiency as evidenced by the 2.5x speedup in Table 3. Altogether, this ablation supports the necessity of a hybrid gating mechanism, one that leverages the locality of GaussianSR and the globality of ContinuousSR.

4.3. Discussion

Takeaways. The experiments demonstrate that our SelectiveSR Gaussian Splatting approach successfully maximizes PSNR on the validation set while maintaining a low memory footprint and a competitive SSIM performance on the validation set. Visual qualitative results, as depicted in Figures 2 and 3 in comparisons of high-frequency textures like butterfly wings and text, convey that our method produces sharper reconstructions than BiCubic interpolation and competitive quality super-resolutions with the remaining baselines.

Although the training time is longer (10 hours) due to the complexity of the gating logic and larger parameter space, the significant reduction in inference time makes the model highly suitable for deployment scenarios where generation speed is critical.

Overall, our work uncovers the importance of a new Gaussian Splatting approach, one that unifies the local precision of a tunable kernel selection strategy as seen in GaussianSR, and the global potency of a consistent prior distribution as used in ContinuousSR. With our results as we have reported them here, image super-resolution efforts can be improved with greater efficiency, producing higher-quality images that can aid in a vast array of computer vision research studies in the future.

Future work. Future work can focus on incorporating textured Gaussian kernels to further improve the sharpness. In addition, further investigations can be made into finetuning the approach on more data to lend it greater versatility, such as the Manga109 dataset [7], a large dataset of different hand-drawn manga drawings. Due to the complexity involved in retrieving and loading the dataset, these results were unable to be included in our final report, and future work can add it for finetuning. Last, another future work can look into incorporating ContinuousSR as another set of ablations, due to its inclusion in our framework. Because of time constraints and unavailability of the training code by the original authors, these results were unable to be included in the final report, and a future study can look into supplying this result.

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Figure 2. Visual comparison of high-frequency texture reconstruction (Butterfly) on Div2K. Our model generates sharp details comparable to SOTA methods.

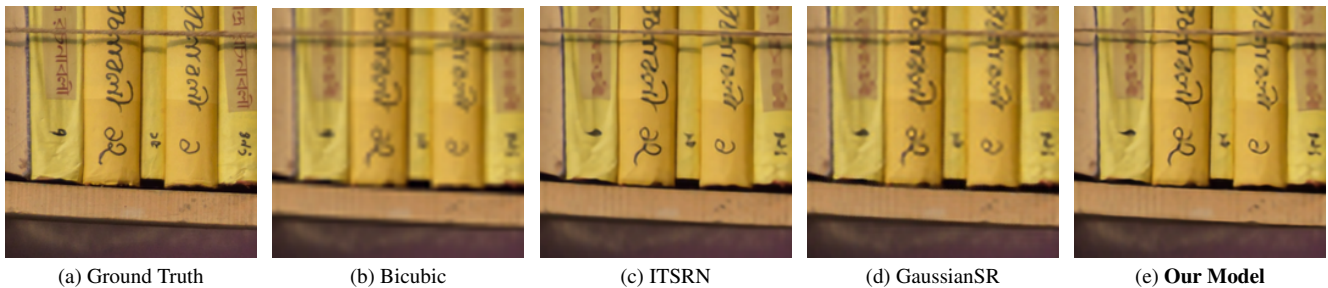


Figure 3. Visual comparison of text legibility (Books). Our model recovers recognizable characters and clearer edges compared to baselines.

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